Sri Chaitanya ACADEMY | Infinity Learn



JEE FINAL TEST SERIES

JEE Advanced Full Syllabus Test - 9 [2024 P1]

Test Date: 04th May 2025

INSTRUCTIONS

- Use the OMR sheet provided separately for answering the questions.
- Blank spaces are provided within this booklet for rough work.
- Write your roll number in the space provided on the front cover of this Question Paper
- You are allowed to take away the Question Paper at the end of the examination.
- The OMR will be collected by the invigilator at the end of the examination.
- Do not tamper with or mutilate the OMR. Do not use the OMR for rough work.
- Write your name, roll number and sign with pen in the space provided for this purpose on the OMR.
- Do not write any of these details anywhere else on the OMR. Darken the appropriate bubble under each digit of your roll number.
- Use a BLACK BALL POINT PEN to darken the bubbles on the OMR>
- The OMR is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- Darken the bubbles ONLY IF you are sure of the answer. There is NO WAY to erase or "un-darken" a darkened bubble.

JEE-ADVANCED-2024-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Matching Type	+3	-1	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Matching Type	+3	-1	4	12
				17	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Matching Type	+3	-1	4	12
Total				**17**	**60**

MATHEMATICS

Max Marks: 60

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. Given that $f(x)+f\left(\frac{1}{1-x}\right)=\frac{2(1-2x)}{x(1-x)}, x\in R, x\neq 0,1$ then the given $f(x)$ is

A) one - one **B)** many one **C)** $f(0)=1$ **D)** none of these

2. Let domain and range of $f(x)$ and $g(x)$ are $[0,\infty)$. If $f(x)$ be an increasing and $g(x)$ be a decreasing function . Also $h(x)=f(g(x)), h(0)=0$ and $p(x)=h\left(x^3-2x^2+2x\right)-h(4)$ then for all x belonging to $(0,2)$

A) $p(x)\in\left[0,-h(4)\right]$ **B)** $p(x)\in\left(-h(4),0\right)$

C) $p(x)\in\left(-h(4),h(4)\right)$ **D)** $p(x)\in\left(h(4),-h(4)\right)$

3. The coefficient $a^8b^4c^9d^9$ in $(abc+abd+acd+bcd)^{10}$

A) 10 ! **B)** $\frac{10!}{8!4!9!9!}$ **C)** 2520 **D)** 2532

4. Total number of ways in which a person can put 8 different rings in the fingers (excluding thumb) of his right hand is equal to

A) ${}^{16}P_8$ **B)** ${}^{11}P_8$ **C)** ${}^{16}C_8$ **D)** ${}^{12}P_8$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. Let $P(x)$ be a polynomial of n degree and $f(x)=\begin{cases}P\left(\frac{1}{x^3}\right)e^{-\frac{1}{x^4}}, & x\neq 0\\ 0, & x=0\end{cases}$, then :

A) $f(x)$ is discontinuous at $x=0$ **B)** $f(x)$ is continuous at $x=0$

C) $f(x)$ is non differentiable at $x=0$ **D)** $f'(0)=\lim\limits_{x\to 0}f(x)$

6. Let p and q be non –zero real numbers. Then the equation $\left(px^2+qy^2+r\right)\left(4x^2+4y^2-8x-4\right)=0$ represents :

A) two straight lines and a circle, when r=0 and p, q are of the opposite sign .

B) two circles, when p=q and r is of sign opposite to that of p .

C) a hyperbola and a circle, when p and q are of opposite sign and $r\neq 0$.

D) a circle and an ellipse, when p and q are unequal but of same sign and r is of sign opposite to that of p .

7. Let $\vec{a}$ and $\vec{b}$ be two non –zero and non-collinear vectors then which of the following is/are always correct ?

A) $\vec{a}\times\vec{b}=\left[\vec{a}\,\vec{b}\,\hat{i}\right]\hat{i}+\left[\vec{a}\,\vec{b}\,\hat{j}\right]\hat{j}+\left[\vec{a}\,\vec{b}\,\hat{k}\right]\hat{k}$

B) $\vec{a}\cdot\vec{b}=\left(\vec{a}\cdot\hat{i}\right)\left(\vec{b}\cdot\hat{i}\right)+\left(\vec{a}\cdot\hat{j}\right)\left(\vec{b}\cdot\hat{j}\right)+\left(\vec{a}\cdot\hat{k}\right)\left(\vec{b}\cdot\hat{k}\right)$

C) If $\vec{u}=\hat{a}-\left(\hat{a}\cdot\hat{b}\right)\hat{b}$ and $\vec{v}=\hat{a}\times\hat{b}$ then $\left|\vec{u}\right|=\left|\vec{v}\right|$

D) If $\vec{c}=\vec{a}\times\left(\vec{a}\times\vec{b}\right)$ and $\vec{d}=\vec{b}\times\left(\vec{a}\times\vec{b}\right)$ then $\vec{c}+\vec{d}=\vec{0}$.

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks :+4 If ONLY the correct integer is entered;Zero Marks : 0 In all other cases.

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8. If $x+\sin y=101$ and $x+101\cos y=100$ where $0\le y\le\frac{\pi}{2}$ then find the value of $[x+y]$.

[Note : [k] denotes the largest integer less than or equal to k]

9. If S_n denotes the sum of n terms of the series 1.2 +2.3+3.4+…. and σ_n denotes the sum of n terms of $\frac{1}{1.2.3.4}+\frac{1}{2.3.4.5}+\frac{1}{3.4.5.6}+\ldots..$, then the value of $18S_n\sigma_{n-1}-S_n+2$ is

10. A circle radius R touches externally a set of 12 circles each of radius r surrounding it. Each one of the smaller circles touches two circles of the set. Then $\frac{R}{r}=\sqrt{m}+\sqrt{n}-1$, where $m,n\in N$ and $m+n$ is

11. Given that z is a non-zero complex number, $iz^2=1+\frac{2}{z}+\frac{3}{z^2}+\frac{4}{z^3}+\frac{5}{z^4}+\ldots..+\infty$ and $z=n\pm\sqrt{-i}$ find the value of 100 n.

12. Let n be a positive integer chosen from the set $\{3,4,5,6,.....,100\}$. $A=\left[a_{ij}\right](1\le i,j\le n)$ be a $n\times n$ matrix such that $a_{ij}\in\{-1,1\}\ \ \forall\ 1\le i,j\le n$. Suppose that $a_{k1}=1\,\forall\,1\le k\le n$ and $\sum_{k=1}^{n}a_{ki}a_{kj}=0$ for all $i\ne j$. If ' m ' be the number of possible value (s) of ' n ' so that the $n\times n$ matrix A can satisfy the above condition, then $\left[\sqrt{m}\right]=$ ____ ([.] denotes G.I.F)

13. Of the three independent events E_1 , E_2 and E_3 the probability that only E_1 occurs is α , only E_2 occurs is β and only E_3 occurs is γ . Let the probability p that none of events E_1, E_2 or E_3 occurs satisfy the equations $(\alpha-2\beta)p=\alpha\beta$ and $(\beta-3\gamma)p=2\beta\gamma$. All the given probabilities are assumed to lie in the interval $(0,1)$ then

$$\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3}=-----$$

SECTION – IV
(MATCHING TYPE)

This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists :**List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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14. Match the statements of Column – I with values of column –II:

	COLUMN-I		**COLUMN-II**
A	If $f(x)=\begin{cases}x+1, when\, x<0\\ x^2-1, when\, x\ge 0\end{cases}$, then $fof(x)$ for $-1\le x<0$ is	P)	$\frac{x-3}{2}$
B	If $f\left(\frac{2\tan x}{1+\tan^2 x}\right)=\frac{(\cos 2x+1)(\sec^2 x+2\tan x)}{2}$ Then $f(x)$ is	Q)	x^2+2x
C	If $f(x+y+1)=\left(\sqrt{f(x)}+\sqrt{f(y)}\right)^2$ for all $x,y\in R$ and $f(0)=1$, then $f(x)$ is	R)	$1+x$
D	If $4<x<5$ and $f(x)=\left[\frac{x}{4}\right]+2x+2$, where $[y]$ Is the greatest integer $\le y$, then $f^{-1}(x)$ is	S)	$(x+1)^2$

A) A-Q;B-R;C-S;D-P **B)** A-P;B-S;C-R;D-Q

C) A-Q;B-S;C-R;D-P D) A-P;B-Q;C-R;D-S

15. Let A,B,C,D be non singular matrices of order 3x3 such that $\left(AA^T\right)BC^2 = B^T$ and $C^2 = I$ and D is an orthogonal matrix such that A^T and D commute with each other and $D^T A^T D = A^{-1}$, then

	COLUMN-I		COLUMN-II
A)	If $B = B^{-1}$, then $\left\|B^{2018}C\right\|$ can be equal to	P)	-1
B)	If $\left(ACA^T\right)^{2019} = AC^K A^T$ then k can be equal to	Q)	0
C)	$\left\|ACA^T\right\|$ can be equal to	R)	2
D)	$\left\|BB^T\left(B^{-1}\right)^2\right\| + \left\|\left(AA^T\right)^{2018}\right\|$ is equal to	S)	3
			4

A) A-S, B-P, C-R, D-P **B)** A-P, B-S, C-P, D-R

C) A-P, B-R, C-S, D-P **D)** A-P, B-S, C-R, D-R

16.

	COLUMN-I		COLUMN-II
A)	The solution of DE $\left(1+x^2y^2\right)ydx + \left(x^2y^2-1\right)xdy = 0$ is	P)	$2ye^{2x} = ce^{2x} - 1$
B)	The solution of DE $2x^3ydy + \left(1-y^2\right)\left(x^2y^2+y^2-1\right)dx = 0$ is	Q)	$4e^{3x} + 3e^{-4y} = c$
C)	The solution of DE $\dfrac{x + \frac{x^3}{3!} + \frac{x^3}{5!} + \ldots}{1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \ldots} = \dfrac{dx-dy}{dx+dy}$ is	R)	$x^2y^2 = 2\ln\left(\dfrac{y}{x}\right) + c$
D)	The solution of $\ln\left(\dfrac{dy}{dx}\right) = 3x + 4y$ is	S)	$x^2y^2 = (cx-1)\left(1-y^2\right)$

A) A-R, B-S, C-Q, D-P **B)** A-R, B-Q, C-P, D-S

C) A-R, B-S, C-P, D-Q **D)** A-R, B-R, C-S, D-Q

17.

	COLUMN-I		COLUMN-II
A)	Area bounded by $\|x\| - \|y\| \geq 1 \& \|x\| \leq 2$ is	P)	8
B)	If $b \in [0,3]$ then the maximum bounded area by curve $y = 1 - \|x-1\| \& y = \|x-b\|$ is	Q)	1/2

C)	Area enclosed by $\left[\lvert x\rvert\right]\left[\lvert y\rvert\right]=2,\left([.]\textit{is G.I.F}\right)$	R)	1
D)	Area bounded $y=\max\left\{\lvert x-2\rvert+2,3-\lvert x-2\rvert\right\}$ & $y=\min\left\{\lvert x-2\rvert+2,3-\lvert x-2\rvert\right\}$	S)	2

A) A-S B-Q, C-P, D-Q

B) A-S, B-R, C-Q, D-P

C) A-S, B-P, C-R, D-Q

D) A-S, B-Q, C-P, D-R

PHYSICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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18. A body travel uniformly a distance of $(s \pm \Delta s)$ in a time $(t \pm \Delta t)$. what may be the condition so that body within the error limits move with a velocity $\left(\frac{s}{t} \pm \frac{\Delta s}{\Delta t}\right)$?

A) $\frac{\Delta t}{t} + \frac{s(\Delta t)^2}{(\Delta s)t^2} = \pm 1$ **B)** $\frac{\Delta t}{t} + \frac{s(\Delta t)}{(\Delta s)t} = \pm 1$

C) $\frac{\Delta t}{t} + \frac{(\Delta s)t}{(\Delta t)s} = \pm 1$ **D)** $\frac{\Delta t}{t} + \frac{s^2(\Delta t)}{(\Delta s)^2 t} = \pm 1$

19. A current i flows through a wire shaped in the form of an infinite parabola of Latus rectum 4a , then magnetic field at the focus of the parabola is

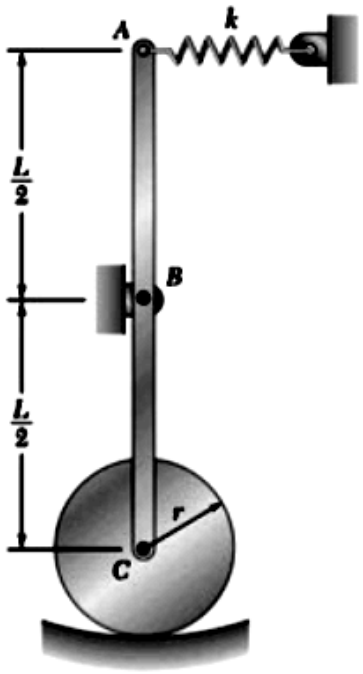
A) $B = \frac{\mu_0 i}{2a}$ **B)** $B = \frac{\mu_0 i}{8a}$ **C)** $B = \frac{\mu_0 i}{4a}$ **D)** $B = \frac{\mu_0 i}{a}$

20. A uniformly charged ring of mass M_0 and radius R_0 is placed on a smooth horizontal surface, The total charge on the ring is Q_0 . A small point charge of mass m_0 and charge q_0 is placed at the centre of the ring. The particle is displaced by a very small distance of $x(x << R_0)$ from the centre in the plane of the ring and released, The ring is not fixed and free to move on the horizontal surface. Find the time after which particle returns to the centre again

A) $T = 2\pi\sqrt{\frac{M_0 m_0 . 2R_0^3}{(M_0 + m_0)4\pi \in_0 Q_0 q_0}}$ **B)** $T = \pi\sqrt{\frac{M_0 m_0 . 8\pi \in_0 R_0^3}{(M_0 + m_0)Q_0 q_0}}$

C) $T = \frac{\pi}{4}\sqrt{\frac{M_0 m_0 . 2R_0^3}{(M_0 + m_0)\pi \in_0 Q_0 q_0}}$ **D)** $T = 2\pi\sqrt{\frac{M_0 m_0 . 2\pi \in_0 R_0^3}{(M_0 + m_0)Q_0 q_0}}$

21. A uniform disc with radius r and mass m can roll without slipping on a cylindrical surface and is attached to a bar ABC with length L and negligible mass. The bar is attached at a point A to a spring with a constant K and can rotate freely about point B in the vertical plane. knowing that end A is given a small displacement and released. Then the frequency of the resulting vibration in terms of m,L,k and g is



A) $f = \dfrac{\sqrt{\dfrac{4k}{3m} + \dfrac{2g}{3L}}}{2\pi}$

B) $f = \dfrac{\sqrt{\dfrac{2k}{3m} + \dfrac{4g}{3L}}}{2\pi}$

C) $f = \dfrac{\sqrt{\dfrac{2k}{3m} + \dfrac{4g}{L}}}{2\pi}$

D) $f = \dfrac{\sqrt{\dfrac{2k}{m} + \dfrac{4g}{3L}}}{\pi}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. An x-ray tube is operating at 50 KV and 20 mA. The target material of the tube has a mass of 1 Kg and specific heat of $495 Jkg^{-1}\,{}^{0}C^{-1}$. one percent of the supplied electric power is converted into x-rays and entire remaining energy goes into heating the target. Then

A) The average rate of rise of temperature of the target would be $2^{0}C$ / sec

B) The minimum wavelength of x-ray emitted is about $0.25\ X\ 10^{-10}\ m$

C) A suitable target material must have high melting temperature.

D) A suitable target material must have low thermal conductivity

23. Two loudspeakers A and B are located 20 m apart and emit sound of frequencies 118 Hz and 121Hz respectively. A car moves along the perpendicular bisector PQR of line AB

where Q is the midpoint of AB. Points P and R lie on opposite sides of Q at a distance of 1800 m from Q. The car starts from point P, crosses Q and then reaches R. Let f(t) represent the beat frequency measured by a person sitting in the car. If the car moves with a uniform speed 60 km/h and the speed of sound is 330 m/s, then the correct statements from the following are

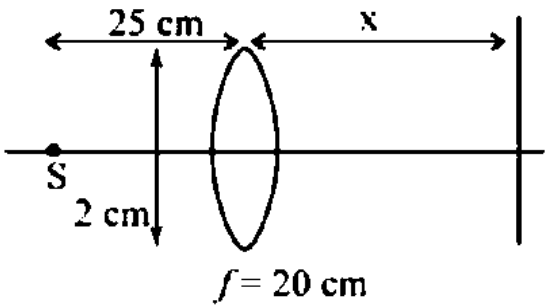
A) Rate of change of f is greatest when the car crosses Q

B) Rate of change of f is least when the car crosses Q

C) $f_P + f_R$ equals four times f_Q

D) $f_P - f_R$ approximately equals one tenth of f_Q

24. A point source of light S is placed on the axis of a thin lens of focal length 20 cm as shown. A screen is placed normal to the axis of lens at a distance X from it. Treat all rays as paraxial.



A) As x is increased from zero , intensity continuously decreases

B) As x is increased from zero ,intensity first increases then decreases

C) Intensity at centre of screen for X=90 cm and X= 110 cm is same

D) Radius of bright circle obtained on screen is equal to 1 cm for X = 200 cm

SECTION-III

(NON-NEGATIVE INTEGER.)

This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.

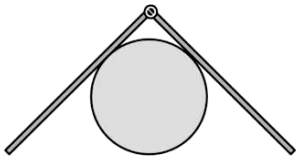
•Answer to each question will be evaluated according to the following marking scheme

Full Marks :+4 If ONLY the correct integer is entered;Zero Marks : 0 In all other cases..

25. A gas consisting of rigid diatomic molecules was expanded in a poly trophic process so that the rate of collisions of the molecules against the vessel's wall did not change. Find the molar heat capacity of the gas in this process is found to be XR. The value of X is_____

26. A Structure consisting of two identical square plates of side $l = 40\ cm$ hinged to a common axle at one of their edges is placed on a fixed horizontal cylinder of radius r =10

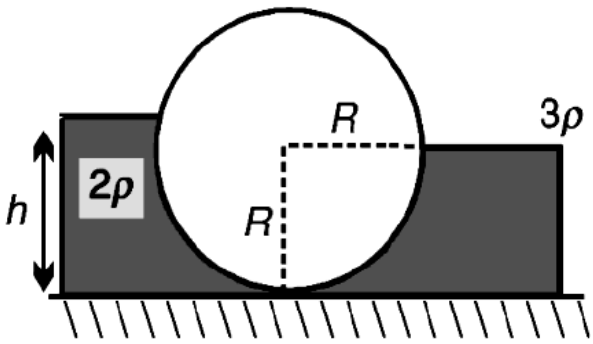
cm as shown in the figure. There is no friction between the cylinder and the plates. At what angle (in degree) with vertical will the plates lean in equilibrium?



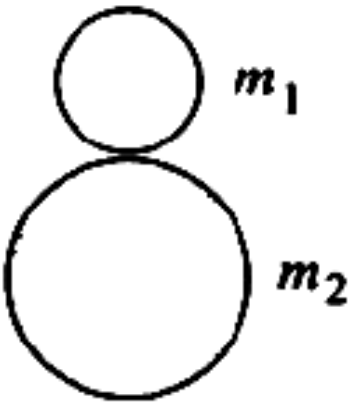
27. Two coherent points sources, S_1 and S_2 vibrating in phase emit light of wavelength λ. The separation between the sources is 3λ. Consider a line passing through S_2 and perpendicular to the line S_1S_2. If the smallest distance from S_2 where a minimum of intensity occurs is $\left(\frac{n+5}{20}\right)\lambda$. find n

28. A man can throw a ball at a speed on the Earth, which can cross a river of width 10m. The man reaches on an imaginary planet whose mean density is twice that of the Earth. Find out the maximum possible radius of the planet(in kilometre) so that if the man throws the ball at the same speed it may escape from the planet. Given radius of earth is $6.4\times10^6 m$

29. In the figure shown, the heavy cylinder $\left(radius\ \sqrt{\frac{8}{3}}m\right)$ kept on a smooth surface separates two liquids of densities 2ρ and 3ρ. The height h(in metre) for the translation equilibrium of the cylinder must be.



30. Two elastic balls of masses m_1 and m_2 are placed on top of each other (with a small gap between them) and then drop onto the ground. The ratio of $m_1\ to\ m_2\ is\ \frac{1}{x}$ for which the upper ball ultimately receives the largest possible fraction of the total energy. Then find the value of x?



SECTION – IV
(MATCHING TYPE)

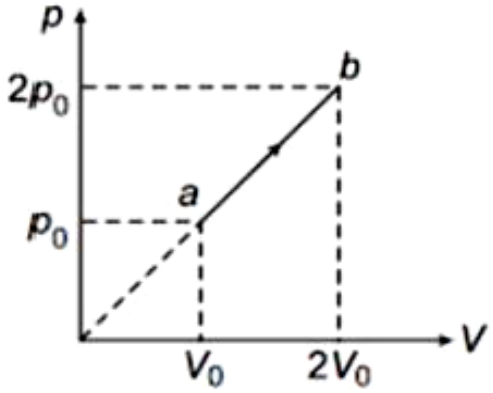
This section contains **FOUR (04)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
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- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
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Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

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31. Heat taken by a gas in process $a \to b$ is $6P_0V_0$ .Match the following columns.



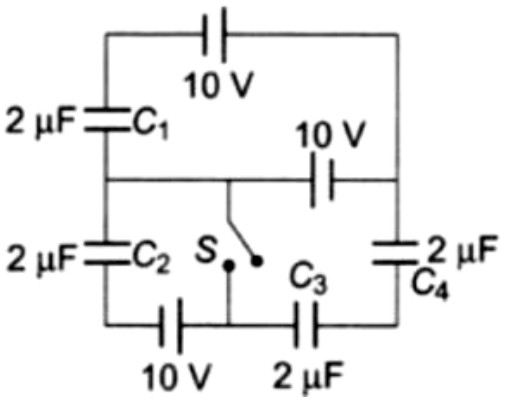
Column I	Column II
(a) W_{ab}	(p) $2P_0V_0$
(b) ΔU_{ab}	(q) $4P_0V_0$
(c) Molar heat capacity in given process	(r) $2R$
(d) C_v of gas	(s) None of these

A) $a \to$s ,b$\to s$,$c \to r$, $d \to s$ **B)** $a \to$s ,b$\to q$,$c \to r$, $d \to s$

C) $a \to$s ,b$\to p$,$c \to s$, $d \to r$ **D)** $a \to$q ,b$\to s$,$c \to r$, $d \to s$.

32. Before connecting the circuit shown in figure, all capacitors were uncharged. The circuit now is in steady state.

Now switch S is closed. During the time, the study state is reached again, match the following:

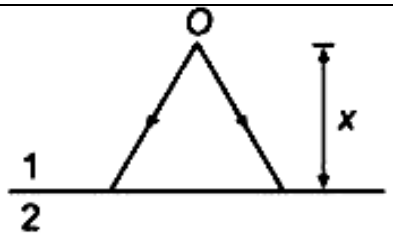
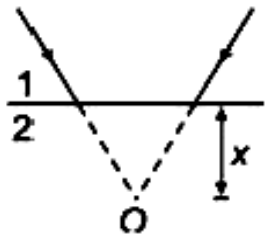
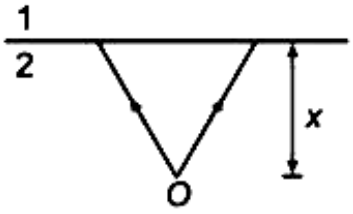


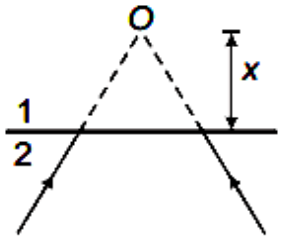
Column I	Column II
a) charge on c_1	p) increases
b)charge on c_2	q) decreases
c)charge on c_3	r) remains same
d)charge on c_4	s) becomes zero

A) $a\rightarrow$r , b$\rightarrow s$,$c\rightarrow r$,$d\rightarrow q$ **B)** $a\rightarrow s$,$b\rightarrow p$,$c\rightarrow r$,$d\rightarrow s$

C) $a\rightarrow$r ,b$\rightarrow p$,$c\rightarrow q$,$d\rightarrow q$ **D)** $a\rightarrow q$,$b\rightarrow s$,$c\rightarrow r$,$d\rightarrow s$

33. Match the following two columns corresponding to single refraction from plane surface . In all cases shown in column I, $\mu_1>\mu_2$

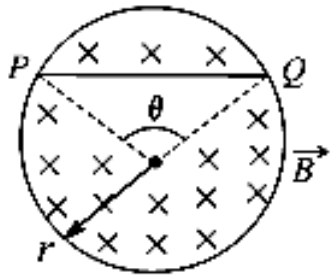
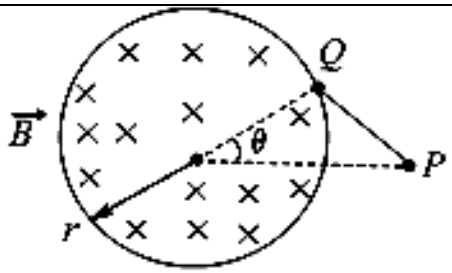
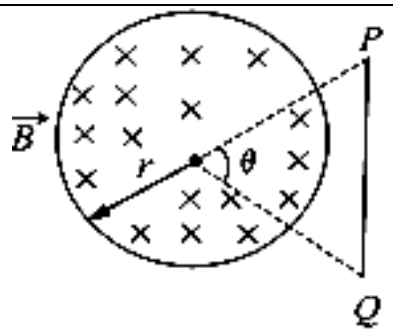
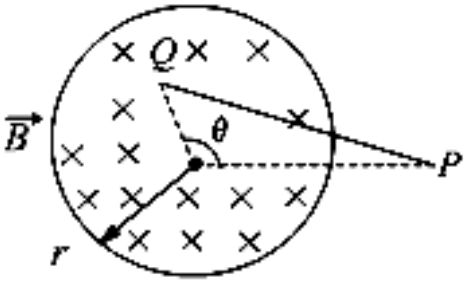
Column I	Column II
a) 	p) Image distance greater than x from plane surface
b) 	q) Image distance less than x from plane surface
c) 	r) Real image

d) 	s) Virtual image

A) $a\rightarrow q,s;\ b\rightarrow q,r;\ c\rightarrow p,s;\ d\rightarrow p$,r **B)** $a\rightarrow q,s;\ b\rightarrow q$,r c$\rightarrow p,r;\ d\rightarrow p,s$

C) $a\rightarrow q$,r; b$\rightarrow q,s;\ c\rightarrow p,s;\ d\rightarrow p,s$ **D)** $a\rightarrow p$,s; b$\rightarrow q,r$; $c\rightarrow p,s;\ d\rightarrow q,r$

34. Column I shows the cylindrical region of radius r where a downward magnetic field $\vec{B}$ exists, where $\vec{B}$ is increasing at the rate of $\frac{d\ B}{dt}$. A rod PQ is placed in different citation as shown. Match the column I with the correct statement in column II regarding the induced emf in the rod.

Column I	Column II
i. 	a. Induced emf in rod PQ is $\frac{1}{2}r^2\theta\frac{dB}{dt}$.
ii. 	b. Induced emf in rod PQ is less than $\frac{1}{2}r^2\theta\frac{dB}{dt}$.
iii. 	c. End P is positive with respect to point Q.
iv. 	d. End Q is positive with respect to point P.

A) $i\rightarrow b,c;ii\rightarrow a,c;\ iii\rightarrow a,d;\ iv\rightarrow b,d$ **B)** $i\rightarrow b$,c; ii $\rightarrow a,d;\ iii\rightarrow a,c;iv\rightarrow b,d$

C) $i \rightarrow b,c$; $ii \rightarrow a,c$; $iii \rightarrow a,d$; $iv \rightarrow b,c$ **D)** $i \rightarrow b$,d; ii$\rightarrow a,c$; $iii \rightarrow a,d$; $iv \rightarrow b,c$

CHEMISTRY **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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35. The appropriate sequence of reagents to produce 2- phenyl but anoic acid from benzene is:

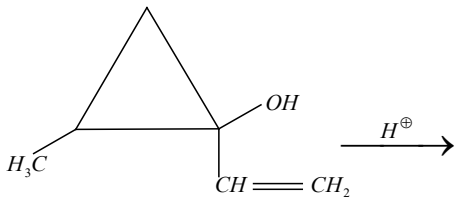
A) (i) 2-chlorobutane/ $AlCl_3$ (ii) NBS, light (iii) $aq.NaCN$ (iv) $H^{\oplus}, H_2O,$ heat

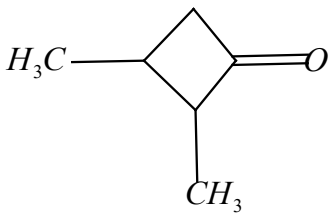
B) (i) 2-chlorobutane/ $AlCl_3$ (ii) $K_2Cr_2O_7 / H_2SO_4$

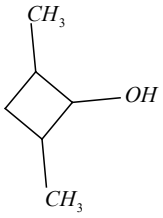
C) (i) propanoyl chloride/ $AlCl_3$ (ii) $Zn - Hg / HCl$ (iii) NBS, light (iv) $NaCN$ (v) $H^{\oplus}, H_2O,$ heat.

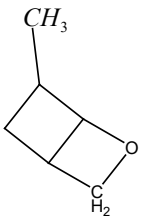
D) i) Butanoyl chloride/ $AlCl_3$ ii) $NaBH_4$ iii) $CuCN$ iv) $H^{\oplus}, H_2O,$ heat

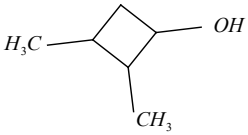
36. The Major Product of the following reaction is



A)


B)


C)


D)


37. Which of the following radicals can decolourise acidified $KMnO_4$ without evolution of any gas

A) SO_3^{-2} **B)** NO_2^- **C)** S^{-2} **D)** All of these

38. Two liquids A and B have the same molecular weight and form and ideal solution the solution is distilled without reflux ¾ of the solution is collected as condensate. In the condensate mole fraction of A = 0.75 and in the reside mole fraction of A = 0 . 3 the mole fraction of 'A' in the original mixture is

A) 0.6375 **B)** 0.3625 **C)** 0.45 **D)** 0.1125

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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39. Which statement is are correct?

A) Benzaldehyde and Aniline gives coupling reaction with diazonium salt

B) Phenol Couples with diazonium salt in mild basic conditions (pH = 8 – 10)

C) Aniline Couples with diazonium salt in mild acidic condition (pH = 4 – 6)

D) $\beta-$ naphthol couples with diazonium salt in neutral condition (pH = 7)

40. In which of the following compounds having general formula of X_2Y_6 there is no X–X bond

A) B_2H_6 **B)** C_2H_6 **C)** I_2Cl_6 **D)** Al_2Cl_6

41. Two test tubes I and II contain solutions of sodium salts of halides in water when Br_2 was added to the solutions the following observations are noted

Test tube	Observation
I	Violet Vapour Emerged
II	No Reaction occurred

If the halides in the tubes I and II are X^- and Y^- respectively and their molecular formula are X_2 and Y_2 respectively the true statements are

A) $S.R.P\ of\ Br_2 > S.R.P\ of\ X_2$ **B)** $S.R.P\ of\ Br_2 > S.R.P\ of\ Y_2$

C) $Y_2\ can\ oxidise\ X^-\ into\ X_2$ **D)** $Y_2\ can\ oxidise\ Br^-\ into\ Br_2$

SECTION-III
(NON-NEGATIVE INTEGER.)

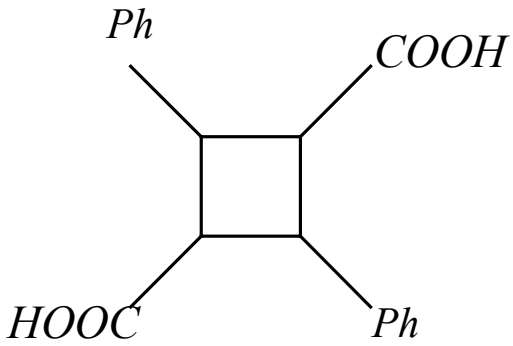
This section contains **SIX (06)** questions.

- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreenvirtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

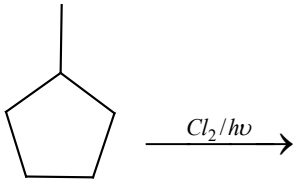
Full Marks :+4 If ONLY the correct integer is entered;Zero Marks : 0 In all other cases.

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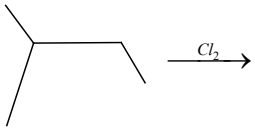
42. How many Geometrical Isomers are possible for the following compound?



43. Consider the following reactions



Total number of monochlorinated product = X (Excluding stereoisomers)



Total number of mono chlorinated product = Y (Excluding stereoisomers)

Identify value of X + Y

44. For how many of the following compound's weight increases in the applied magnetic field

$NO, NO_2, O_2, KO_2, MnSO_4, NiSO_4 CuSO_4, ZnSO_4$

45. When 0.03 lit of a mixture of hydrogen and oxygen was exploded 0.003 lit of oxygen remained. The Initial mixture contains $10x\%\ of\ O_2$ The value of X is (All volumes are measured at same temperature and pressure)

46. Percentage of void space in AB solid having rock salt structure is $10x$ Given that $\frac{r+}{r-} = \frac{1}{2}$ having cation anion contact and $\pi = 3.15$ the value of x is

47. Given that value of vanderwaal's constant for CO_2 gas as a = 3.592 dm^6 atm mole and volume occupied by a CO_2 molecule is negligible the pressure exerted by one mole of CO_2 at $273k$ mole of CO_2 is 5X. The value of 'X' is adjust to the nearest integer.

SECTION – IV
(MATCHING TYPE)

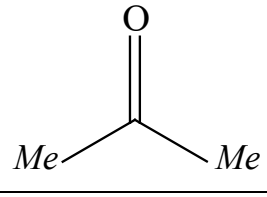
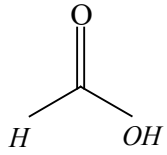
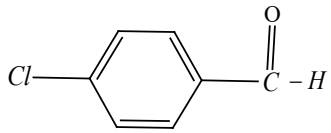
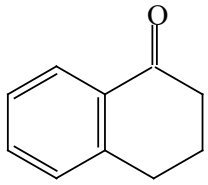
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Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

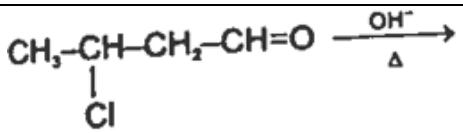
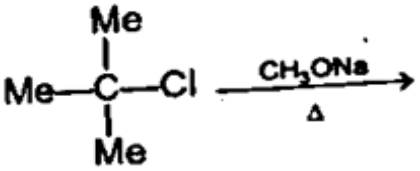
Negative Marks: –1 In all other cases..

48. In column –I some compounds are given,? In column –II some reagent given. Match the compounds in the Column I with Column II

Column – I	**Column – II**
A) 	p) 2,4 – D N P test
B) 	q) Tollen's Reagent test
C) 	r) I_2 + NaOH (Iodoform test)
D) 	s) Fehlings Solutions test

A) A-p, s; B - q, r; C - p, q; D - p **B)** A - p; B - p, q; C - q, s; D - p,r
C) A - p, r; B - q, s; C - p, q; D – p **D)** A = q,s; B - p,r; C - p, q; D - p

49. Match the Column
Match List – I (Compounds) with their most appropriate mechanism (List-II)

Column – I	**Column – II**
A) 	(p) E2
B) $CH_3 - CH_2 - I \xrightarrow{OH^-}$	(q) E1cB
C) 	(r) S_N2
D) $CH_3 - CH_2 - CH_2 - O - Ph \xrightarrow{Conc.HI}$	(s) Intermediate carbanion
	(t) Dueterium Exchange in presence of OD^- / D_2O

A) A = q, st, B = r, C = p , D =r **B)** A = r , B = q,s,t C = p, D = s
C) A = st, B = q, C = r, D = p **D)** A = s,t,r B = p C = q D = r

50. Matching the following

Column – I	**Column – II**

P) O_2 *and* NO^-	1) Same Magnetic Property and bond order as that in N_2^+
Q) O_2^+ *and* NO	2) Same bond order but not same magnetic property as that O_2
R) CO *and* CN^-	3) Same magnetic property and bond order as that N_2^{-2}
S) 4) C_2 *and* CN^+	4) Same magnetic property and bond order as that NO^+

The correct combination

A) P-2;Q-4;R-1;S- 3 **B)** P-4;Q-1 ; R-2 ; S-3

C) P-3; Q-2; R-1 ; S-4 **D)** P-4; Q-2; R-3; S-1

51. 0.1 Mole of each solute in the List – I are dissolved in 10 moles of water separately

List – I	**List – II**
P) $AlCl_3$ if $\alpha = 0.8$	1) $i = 3.4$
Q) $BaCl_2$ is $\alpha = 0.9$	2) Has minimum osmotic pressure among the solutions
R) Na_3PO_4 is $\alpha = 0.9$	3) Has minimum freezing point among the given solutions
S) $K_4\left[Fe(CN)_6\right]$ if $\alpha = 0.7$	4) has $RLVP = \frac{37}{1037}$

A) P $\rightarrow$ 4;Q $\rightarrow$ 2; R $\rightarrow$ 3; S $\rightarrow$ 1 **B)** P $\rightarrow$ 1;Q $\rightarrow$ 2; R $\rightarrow$ 4; S $\rightarrow$ 3

C) P $\rightarrow$ 2;Q $\rightarrow$ 1; R $\rightarrow$ 3; S $\rightarrow$ 4 **D)** P $\rightarrow$ 1;Q $\rightarrow$ 2; R $\rightarrow$ 3; S $\rightarrow$ 4